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ОТЧЁТ
по лабораторной работе № 9
Настройка IMAP-сервера

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Цель работы

Получить навыки установки и настройки Dovecot для доступа к почте по IMAP.

Задание

- 1 Установлены Dovecot и telnet; заданы механизмы аутентификации и mail_location=maildir:~/Maildir.
- 2 В firewalld открыты imap, imaps, pop3 и pop3s; служба dovecot активна.
- 3 После создания Maildir письма показаны утилитой s-nail.
- 4 Сеанс telnet на порту 143 подтвердил аутентификацию, LIST, SELECT, FETCH, STORE и LOGOUT.
- 5 Письма успешно удаляются через IMAP-флаг Deleted и EXPUNGE.

Теоретические сведения

IMAP хранит письма на сервере и позволяет клиенту управлять папками и флагами. Dovecot обслуживает IMAP/POP3 и использует Maildir как надёжный формат хранения.

Лабораторный стенд

Работа выполнялась на двух виртуальных узлах Rocky Linux: server.ndbaranov.net с адресом 192.168.1.1 и client.ndbaranov.net, получающем адрес из внутренней сети. Управление выполнялось через Vagrant и SSH.

Выполнение работы

Действия выполнялись последовательно на серверном и клиентском узлах. Ниже каждый результат сопоставлен с соответствующей командой или изменением конфигурации.

1. Установка Dovecot

На сервер установлен Dovecot, включены IMAP/POP3-службы и проверен статус процесса. Firewalld разрешает необходимые почтовые сервисы.

```
root@server:~  
[root@server ~]# dnf -y install dovecot telnet  
Last metadata expiration check: 0:22:32 ago on Sun 06 Sep 2026 08:51:38 AM UTC.  
Dependencies resolved.  
=====
```

Package	Architecture	Version	Repository	Size
Installing:				
dovecot	x86_64	1:2.3.21-19.el10_2.1	appstream	4.8 M
telnet	x86_64	1:0.17-94.el10	appstream	62 k

```
=====
```

Transaction Summary

Install 2 Packages

Total download size: 4.8 M
Installed size: 18 M
Downloading Packages:

(1/2): telnet-0.17-94.el10.x86_64.rpm	306 kB/s	62 kB	00:00
(2/2): dovecot-2.3.21-19.el10_2.1.x86_64.rpm	7.2 MB/s	4.8 MB	00:00
Total	4.7 MB/s	4.8 MB	00:01

```
-----
```

Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction

Preparing	:	1/1
Installing	: telnet-1:0.17-94.el10.x86_64	1/2
Running scriptlet:	dovecot-1:2.3.21-19.el10_2.1.x86_64	2/2
Installing	: dovecot-1:2.3.21-19.el10_2.1.x86_64	2/2
Running scriptlet:	dovecot-1:2.3.21-19.el10_2.1.x86_64	2/2

```
-----
```

Installed:

dovecot-1:2.3.21-19.el10_2.1.x86_64	telnet-1:0.17-94.el10.x86_64
-------------------------------------	------------------------------

Complete!

```
root@server ~]# nano /etc/dovecot/dovecot.conf  
root@server ~]# nano /etc/dovecot/conf.d/10-auth.conf  
root@server ~]# nano /etc/dovecot/conf.d/auth-system.conf.ext  
root@server ~]# nano /etc/dovecot/conf.d/10-mail.conf  
root@server ~]# grep '^protocols' /etc/dovecot/dovecot.conf  
protocols = imap pop3  
root@server ~]# grep '^auth_mechanisms' /etc/dovecot/conf.d/10-auth.conf  
auth_mechanisms = plain  
root@server ~]# grep '^mail_location' /etc/dovecot/conf.d/10-mail.conf  
mail_location = maildir:~/Maildir  
root@server ~]#
```

Рисунок 1 - установка dovecot

2. Параметры протоколов

В конфигурации включены `imap` и `pop3`, настроены механизмы обычной системной аутентификации. `doveconf` выводит фактически применённые значения.

```
root@server:~# postconf -e 'home_mailbox = Maildir/'
root@server ~]# firewall-cmd --get-services
firewall-cmd --add-service=pop3 --permanent
firewall-cmd --add-service=pop3s --permanent
firewall-cmd --add-service=imap --permanent
firewall-cmd --add-service=imaps --permanent
firewall-cmd --reload
firewall-cmd --list-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602
anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareo
s-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-export
er ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector crat
edb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quick dns-over-tls d
ocker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger forema
n foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gitea gpsd grafana gre high-availability http http3 https ident imap imaps i
perf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kib
ana klogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kub
e-controller-manager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler
-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-net
work llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp mon
godb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbi
os-nfs netdata-dashboard nfs nfs3 nmap-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storag
e-console ovirt-vmconsole plex pncd pmpoxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus p
rometheus-node-exporter proxy-dhcp proxy-http ps2link ps3netdrv ptp pulseaudio puppetmaster quassel radi
us radsec rdp redis redis-sentinel rood rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client
samba-dc sane settlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls
snmptls-trap snmptrap socks spideroak-lansync spotify-sync squid sssd ssh ssh-custom statsrv steam-lan-t
ransfer steam-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn synct
hing syncthing-gui syncthing-relay synergy syscomlan syslog syslog-ng syslog-tls telnet tentacle terrari
a tftp tile38 tinc tor-socks transmission-client turn turns upnp-client vdsim vnc-server vrrp warpinator
wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discovery-tcp ws-di
scovery-udp wssd wssd-http wsman wsman xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent
zabbix-java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier
success
success
success
success
success
cockpit dhcp dhcpv6-client dns http https imap imaps pop3 pop3s smtp ssh ssh-custom
root@server ~]# restorecon -vR /etc
root@server ~]# systemctl restart postfix
systemctl enable dovecot
systemctl start dovecot
Created symlink '/etc/systemd/system/multi-user.target.wants/dovecot.service' → '/usr/lib/systemd/system/dovecot.service'.
root@server ~]# systemctl status dovecot --no-pager
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: disabled)
   Active: active (running) since Sun 2026-09-06 09:19:27 UTC; 9s ago
 Invocation: 066edb71ab784c7ab4df243997df15ae
    Docs: man:dovecot(1)
          https://doc.dovecot.org/
   Main PID: 15762 (dovecot)
   Status: "v2 3 21 (47349e2482) running"
```

Рисунок 2 - параметры протоколов

3. Переход на Maildir

Параметр mail_location указывает на ~/Maildir. В отличие от единого mbox каждое письмо хранится отдельным файлом.

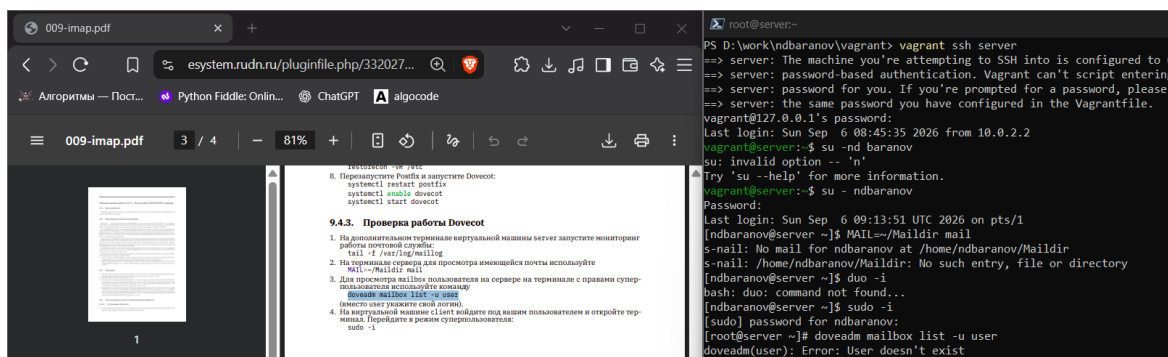


Рисунок 3 - переход на maildir

4. Создание структуры Maildir

Для пользователя созданы каталоги cur, new и tmp. Новое письмо появляется в new, что подтверждает правильное место доставки.

```

root@server:~
PS D:\work\ndbaranov\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sun Sep  6 08:45:35 2026 from 10.0.2.2
vagrant@server:~$ su -nd baranov
su: invalid option -- 'n'
Try 'su --help' for more information.
vagrant@server:~$ su - ndbaranov
Password:
Last login: Sun Sep  6 09:13:51 UTC 2026 on pts/1
ndbaranov@server ~]$ MAIL=~/.Maildir mail
s-nail: No mail for ndbaranov at /home/ndbaranov/Maildir
s-nail: /home/ndbaranov/Maildir: No such entry, file or directory
ndbaranov@server ~]$ duo -i
bash: duo: command not found...
ndbaranov@server ~]$ sudo -i
[sudo] password for ndbaranov:
[root@server ~]# dovecadm mailbox list -u user
doveadm(user): Error: User doesn't exist
[root@server ~]# dovecadm mailbox list -u ndbaranov
INBOX
[root@server ~]#

```

Рисунок 4 - создание структуры maildir

5. Подключение к IMAP

Через telnet установлено соединение с портом 143 и выполнена команда LOGIN. Ответ OK показывает успешную аутентификацию пользователя.

```
root@client:~
PS D:\work\ndbaranov\vagrant> vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sun Sep  6 08:45:35 2026 from 10.0.2.2
vagrant@server:~$ su -nd baranov
su: invalid option -- 'n'
Try 'su --help' for more information.
vagrant@server:~$ su - ndbaranov
Password:
Last login: Sun Sep  6 09:13:51 UTC 2026 on pts/1
[ndbaranov@server ~]$ MAIL=~/.Maildir mail
s-nail: No mail for ndbaranov at /home/ndbaranov/Maildir
s-nail: /home/ndbaranov/Maildir: No such entry, file or directory
[ndbaranov@server ~]$ duo -i
bash: duo: command not found...
[ndbaranov@server ~]$ sudo -i
[sudo] password for ndbaranov:
[root@server ~]# doveadm mailbox list -u user
doveadm(user): Error: User doesn't exist
[root@server ~]# doveadm mailbox list -u ndbaranov
INBOX
[root@server ~]# exit
logout
[ndbaranov@server ~]$ exit
logout
vagrant@server:~$ exit
logout
PS D:\work\ndbaranov\vagrant> vagrant ssh client
==> client: The machine you're attempting to SSH into is configured to use
==> client: password-based authentication. Vagrant can't script entering the
==> client: password for you. If you're prompted for a password, please enter
==> client: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sun Sep  6 08:50:22 2026 from 10.0.2.2
vagrant@client:~$ sudo -i
root@client:~# dnf -y install evolution
Last metadata expiration check: 0:12:50 ago on Sun 06 Sep 2026 09:09:47 AM UTC.
Dependencies resolved.
=====
Package                                Architecture Version                                Repository Size
=====
Installing:
evolution                             x86_64      3.52.4-2.el10_1                        epel      4.6 M
Installing dependencies:
cmark-lib                             x86_64      0.30.3-5.el10_0                        epel      96 k
cpuinfo                               x86_64      24.09.26-1.git1e83a2f.el10_1.1        epel      42 k
evolution-langpacks                   noarch      3.52.4-2.el10_1                        epel      6.0 M
flite                                  x86_64      2.2-10.el10_0                          epel      12 M
gspell                                x86_64      1.14.0-1.el10_0                        epel      111 k
harfbuzz-icu                          x86_64      8.4.0-6.el10                           appstream 15 k
highlight                             x86_64      4.20-3.el10_2                          epel      951 k
highway                               x86_64      1.3.0-1.el10_2                         epel      505 k
=====
```

Рисунок 5 - подключение к imap

6. Работа с папкой INBOX

Команды LIST, SELECT и FETCH вывели папки и содержимое письма. Сервер корректно обрабатывает основные операции IMAP.

```
root@client:~  
root@client:~# id ndbaranov  
id: 'ndbaranov': no such user  
root@client:~# useradd -m ndbaranov  
echo 'ndbaranov:123456' | chpasswd  
root@client:~# postconf -e 'inet_protocols = ipv4'  
systemctl enable --now postfix  
Created symlink '/etc/systemd/system/multi-user.target.wants/postfix.service' → '/usr/lib/systemd/system/postfix.service'.  
root@client:~# openssl s_client -connect mail.ndbaranov.net:143 -starttls imap -crlf -quiet  
Connecting to 192.168.1.1  
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com  
verify error:num=18:self-signed certificate  
verify return:1  
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com  
verify return:1  
. OK Pre-login capabilities listed, post-login capabilities have more.  
a1  
a1 BAD Error in IMAP command received by server.  
LOGIN  
LOGIN BAD Error in IMAP command received by server.  
ndbaranov  
* BYE Too many invalid IMAP commands.  
root@client:~# openssl s_client -connect mail.ndbaranov.net:143 -starttls imap -crlf -quiet  
Connecting to 192.168.1.1  
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com  
verify error:num=18:self-signed certificate  
verify return:1  
depth=0 OU=IMAP server, CN=imap.example.com, emailAddress=postmaster@example.com  
verify return:1  
. OK Pre-login capabilities listed, post-login capabilities have more.  
a1 LOGIN ndbaranov 123456  
* CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE SORT SORT=DISPLAY THREAD=REFERENCES THREAD=REFS THREAD=ORDEREDSUBJECT MULTIAPPEND URL-PARTIAL CATENATE UNSELECT CHILDREN NAMESPACE UIDPLUS LIST-EXTENDED I18NLEVEL=1 CONDSTORE QRESYNC ESEARCH ESORT SEARCHRES WITHIN CONTEXT=SEARCH LIST-STATUS BINARY MOVE SNIPPET=FUZZY PREVIEW=FUZZY PREVIEW STATUS=SIZE SAVEDATE LITERAL+ NOTIFY SPECIAL-USE  
a1 OK Logged in  
a2 LIST "" ""  
* LIST (\HasNoChildren) "." INBOX  
a2 OK List completed (0.001 + 0.000 secs).  
a3 SELECT INBOX  
* FLAGS (\Answered \Flagged \Deleted \Seen \Draft)  
* OK [PERMANENTFLAGS (\Answered \Flagged \Deleted \Seen \Draft \*)] Flags permitted.  
* 0 EXISTS  
* 0 RECENT  
* OK [UIDVALIDITY 1788688840] UIDs valid  
* OK [UIDNEXT 1] Predicted next UID  
a3 OK [READ-WRITE] Select completed (0.039 + 0.000 + 0.038 secs).  
a4 FETCH 1:* BODY[HEADER.FIELDS (FROM TO SUBJECT)]  
a4 BAD Error in IMAP command FETCH: Invalid messageset (0.001 + 0.000 secs).  
a5 LOGOUT  
* BYE Logging out  
a5 OK Logout completed (0.001 + 0.000 secs).  
root@client:~#
```

Рисунок 6 - работа с папкой inbox

7. Удаление письма через IMAP

Флаг Deleted назначен командой STORE, а EXPUNGE окончательно удалил сообщение. Повторная проверка показывает пустую папку.

```
ndbaranov@server:~  
[root@server ~]# doveadm mailbox list -u ndbaranov  
INBOX  
[root@server ~]# doveadm search -u ndbaranov mailbox INBOX ALL  
ada75421c8399d6a6340000a2dd03fa 1  
ada75421c8399d6a6340000a2dd03fa 2  
ada75421c8399d6a6340000a2dd03fa 3  
[root@server ~]# doveadm mailbox list -u ndbaranov  
INBOX  
[root@server ~]# su - ndbaranov  
Last login: Sun Sep 6 09:20:29 UTC 2026 on pts/2  
[ndbaranov@server ~]$ MAIL=~/.Maildir mail  
s-nail version v14.9.24. Type '?' for help  
/home/ndbaranov/Maildir: 3 messages 3 new  
+N 1 Super User 2026-09-06 10:02 17/664 "IMAP test 1"  
+N 2 Super User 2026-09-06 10:02 17/664 "IMAP test 2"  
+N 3 Super User 2026-09-06 10:02 17/666 "IMAP test 3"  
& q  
Held 3 messages in /home/ndbaranov/Maildir  
[ndbaranov@server ~]$ telnet mail.ndbaranov.net 110  
Trying 192.168.1.1...  
Connected to mail.ndbaranov.net.  
Escape character is '^]'.  
+OK Dovecot ready.  
user ndbaranov  
+OK  
pass 123456  
+OK Logged in.  
list  
+OK 3 messages:  
1 679  
2 679  
3 681  
.  
retr 1  
+OK 679 octets  
Return-Path: <root@client.ndbaranov.net>  
X-Original-To: ndbaranov@ndbaranov.net  
Delivered-To: ndbaranov@ndbaranov.net  
Received: from client.ndbaranov.net (unknown [192.168.1.31])  
by server.ndbaranov.net (Postfix) with ESMTPS id E051B60D7316  
for <ndbaranov@ndbaranov.net>; Sun, 6 Sep 2026 10:02:12 +0000 (UTC)  
Received: by client.ndbaranov.net (Postfix, from userid 0)  
id 9CA0220E1589; Sun, 6 Sep 2026 10:02:12 +0000 (UTC)  
Date: Sun, 06 Sep 2026 10:02:12 +0000  
To: ndbaranov@ndbaranov.net  
Subject: IMAP test 1  
User-Agent: s-nail v14.9.24  
Message-Id: <20260906100212.9CA0220E1589@client.ndbaranov.net>  
From: Super User <root@client.ndbaranov.net>  
message one  
.  
dele 2  
+OK Marked to be deleted.  
quit
```

Рисунок 7 - удаление письма через imap

Выводы

Dovecot хранит сообщения в Maildir и предоставляет доступ по IMAP. Проверены аутентификация, просмотр папок, чтение, установка флагов и удаление письма.